

Chapter 10: Analysing Data

Three-week revision booklet · 6 worksheets · 240 questions

NSW syllabus outcomes

MA4-DAT-C-01 (classifying and displaying data) · MA4-DAT-C-02
(measures of centre and range) · MAO-WM-01

NAME _____

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How to use this booklet

There are six worksheets, two for each week. Each one drills a single skill from the Analysing Data topic test. Each sheet has the same shape: **40 questions**, split into ● **Easy**, ● **Medium** and ● **Challenging**. Work from the top of the page down — the questions get harder as you go. Show your working for the challenging questions. The final worksheet mixes every skill together, so it is the best check that you are ready for the topic test.

Week	Worksheet	Skill
1	1	Mean
1	2	Median and mode
2	3	Range and outliers
2	4	Frequency tables and plots
3	5	Reading and interpreting graphs
3	6	Mixed revision — all skills

Progress tracker

Worksheet	Date completed	Score / 40	Checked by
1 Mean			
2 Median and mode			
3 Range and outliers			
4 Frequency tables and plots			
5 Reading and interpreting graphs			
6 Mixed revision — all skills			

Worksheet 1: Mean

Week 1 · Worksheet 1 of 6 · 40 questions

NSW syllabus outcomes

MA4-DAT-C-01 (classifying and displaying data) · MA4-DAT-C-02
(measures of centre and range) · MAO-WM-01

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The mean is the total of all scores divided by how many scores there are.

EASY

20 QUESTIONS

Find the mean of each data set.

- | | |
|----------------------|--------------------------|
| 1. 2, 4, 6 _____ | 2. 5, 5, 8 _____ |
| 3. 1, 2, 3, 4 _____ | 4. 10, 20, 30 _____ |
| 5. 3, 7, 8 _____ | 6. 6, 6, 6, 6 _____ |
| 7. 4, 8, 12 _____ | 8. 2, 3, 4, 7 _____ |
| 9. 9, 11 _____ | 10. 0, 5, 10 _____ |
| 11. 12, 14, 16 _____ | 12. 1, 1, 4 _____ |
| 13. 20, 25, 30 _____ | 14. 7, 9, 11, 13 _____ |
| 15. 2, 8 _____ | 16. 5, 10, 15, 20 _____ |
| 17. 3, 3, 9 _____ | 18. 8, 12, 16, 24 _____ |
| 19. 15, 25 _____ | 20. 6, 7, 8, 9, 10 _____ |

MEDIUM

12 QUESTIONS

Find the mean. Round to 2 decimal places if needed.

- | | |
|----------------------------------|--|
| 21. 5, 2, 3, 6, 4, 5, 8 _____ | 22. 6, 12, 11, 12, 10, 6, 6, 10, 6 _____ |
| 23. 8, 4, 1, 1, 4, 1, 3, 6 _____ | 24. 20, 21, 23, 24, 25 _____ |
| 25. 14, 18, 9, 11 _____ | 26. 3, 3, 4, 5, 5, 6 _____ |
| 27. 100, 150, 125 _____ | 28. 7, 2, 9, 4, 8, 6 _____ |
| 29. 1.5, 2.5, 3.5 _____ | 30. 45, 52, 48, 51 _____ |
| 31. 0, 0, 5, 7 _____ | 32. 13, 17, 19, 23, 28 _____ |

CHALLENGING

8 QUESTIONS

Answer each question. Show your working.

- | | |
|---|---|
| 33. The mean of 4 numbers is 7. Find their total.
..... | 34. The mean of 5 numbers is 12. Four of them are 10, 11, 13 and 15. Find the fifth.
..... |
| 35. A small data set has a mean of 6. An extra value of 20 is added. Is the new mean higher, lower or the same? Explain.
..... | 36. Five test marks average 68. What total was scored?
..... |
| 37. The mean of 3, 7 and x is 6. Find x.
..... | 38. A student scores 15, 18 and 12 out of 20. Find the mean as a percentage.
..... |
| 39. Adding a score equal to the mean does not change the mean. Explain why.
..... | 40. The mean of six numbers is 9. If each number is doubled, what is the new mean?
..... |

Worksheet 2: Median and mode

Week 1 • Worksheet 2 of 6 • 40 questions

NSW syllabus outcomes

MA4-DAT-C-01 (classifying and displaying data) • MA4-DAT-C-02
(measures of centre and range) • MAO-WM-01

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The median is the middle score once the data is ordered. The mode is the score that appears most often.

EASY

20 QUESTIONS

Find the median of each ordered data set.

- | | |
|-------------------------|-----------------------------|
| 1. 1, 3, 5 _____ | 2. 2, 4, 6, 8 _____ |
| 3. 3, 5, 7, 9, 11 _____ | 4. 10, 20, 30, 40 _____ |
| 5. 1, 2, 3, 4, 5 _____ | 6. 6, 8, 10 _____ |
| 7. 4, 4, 9, 9 _____ | 8. 2, 5, 8, 11, 14 _____ |
| 9. 7, 9 _____ | 10. 1, 1, 2, 3, 5, 8 _____ |
| 11. 12, 15, 18 _____ | 12. 20, 22, 24, 26 _____ |
| 13. 0, 1, 1, 4, 9 _____ | 14. 5, 10, 15, 20, 25 _____ |
| 15. 3, 6 _____ | |

Find the mode of each data set.

- | | |
|----------------------------|------------------------|
| 16. 2, 3, 3, 5 _____ | 17. 6, 6, 6, 9 _____ |
| 18. 1, 4, 4, 7, 7, 7 _____ | 19. 8, 9, 10, 10 _____ |
| 20. 5, 5, 2, 3, 2, 5 _____ | |

MEDIUM

12 QUESTIONS

Find the median and the mode.

- | | |
|----------------------------------|--|
| 21. 5, 2, 3, 6, 4, 5, 8 _____ | 22. 6, 12, 11, 12, 10, 6, 6, 10, 6 _____ |
| 23. 8, 4, 1, 1, 4, 1, 3, 6 _____ | 24. 3, 7, 3, 9, 5, 3 _____ |
| 25. 20, 21, 23, 24, 25 _____ | 26. 14, 14, 18, 9, 11 _____ |
| 27. 2, 2, 4, 4, 6 _____ | 28. 35, 36, 36, 37, 37, 37 _____ |
| 29. 1, 5, 9, 5, 3 _____ | 30. 12, 8, 12, 6, 12, 8 _____ |
| 31. 4, 4, 5, 6, 6 _____ | 32. 30, 35, 35, 36, 39, 41 _____ |

CHALLENGING

8 QUESTIONS

Answer each question. Show your working.

- | | |
|--|---|
| 33. A data set is 4, 7, 9, x, 12 in order. If the median is 9, what can x be?
..... | 34. A data set has mode 6 and contains 4, 5, 6, 6, 7. Add one value so the mode stays 6.
..... |
| 35. Can a data set have two modes? Give an example.
..... | 36. Can a data set have no mode? Give an example.
..... |
| 37. The median of 3, 8, x, 15 is 9. Find x.
..... | 38. 21 students' mothers' ages have median 37 and mode 36. Which is more typical? Explain.
..... |
| 39. Explain why the median is not affected by one very large value.
..... | 40. For 2, 4, 6, 8, 10, find the mean, median and mode, or state if there is none.
..... |

Worksheet 3: Range and outliers

Week 2 • Worksheet 3 of 6 • 40 questions

NSW syllabus outcomes

MA4-DAT-C-01 (classifying and displaying data) • MA4-DAT-C-02
(measures of centre and range) • MAO-WM-01

NAME _____

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The range is the largest score minus the smallest. An outlier sits well away from the rest.

EASY

20 QUESTIONS

Find the range of each data set.

- | | |
|----------------------------|--------------------------|
| 1. 3, 7, 9 _____ | 2. 2, 8 _____ |
| 3. 10, 15, 20, 25 _____ | 4. 1, 1, 1 _____ |
| 5. 4, 9, 16, 25 _____ | 6. 0, 100 _____ |
| 7. 12, 14, 19 _____ | 8. 5, 5, 8, 12 _____ |
| 9. 20, 35, 27 _____ | 10. 6, 6, 7, 9 _____ |
| 11. 45, 52, 48 _____ | 12. 2.5, 4.5 _____ |
| 13. 7, 13, 11, 9 _____ | 14. 100, 250, 175 _____ |
| 15. 8, 8, 15, 3 _____ | 16. 30, 22, 41, 19 _____ |
| 17. 1, 2, 3, 4, 5, 6 _____ | 18. 18, 18 _____ |
| 19. 9, 21, 15, 12 _____ | 20. 50, 65, 58, 72 _____ |

MEDIUM

12 QUESTIONS

Find the range, and name the outlier if there is one.

- | | |
|-----------------------------|------------------------------|
| 21. 3, 4, 5, 6, 40 _____ | 22. 12, 13, 14, 15 _____ |
| 23. 1, 2, 2, 3, 25 _____ | 24. 50, 52, 51, 8 _____ |
| 25. 6, 7, 7, 8, 9 _____ | 26. 100, 102, 101, 5 _____ |
| 27. 0, 1, 2, 3, 4, 30 _____ | 28. 22, 24, 23, 25, 60 _____ |
| 29. 5, 6, 7, 8, 9, 10 _____ | 30. 2, 3, 3, 4, 4, 19 _____ |
| 31. 31, 33, 32, 34, 2 _____ | 32. 9, 10, 11, 12, 13 _____ |

CHALLENGING

8 QUESTIONS

Answer each question. Show your working.

- | | |
|--|--|
| 33. A dot plot shows goals 0, 1, 1, 2, 2, 3, 3, 3, 4, 4, 7. Find the range and the outlier.
..... | 34. For that same data, find the mode and the median.
..... |
| 35. In how many matches did the team score 2 goals?
..... | 36. A data set has range 18 and smallest value 7. Find the largest value.
..... |
| 37. A data set has range 25 and largest value 40. Find the smallest value.
..... | 38. Explain how an outlier affects the mean but not the median.
..... |
| 39. One value of 39 in a set is corrected to 40. Describe the effect on mean, median, mode and range.
..... | 40. Give a data set of five numbers with mean 10 and range 8.
..... |

Worksheet 4: Frequency tables and plots

Week 2 • Worksheet 4 of 6 • 40 questions

NSW syllabus outcomes

MA4-DAT-C-01 (classifying and displaying data) • MA4-DAT-C-02
(measures of centre and range) • MAO-WM-01

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In a frequency table, fx means score $\times$ frequency. The total of fx divided by the total frequency gives the mean.

EASY

20 QUESTIONS

A stem-and-leaf plot has stems 2 | 8 3 | 2 5 9 4 | 0 1 2 3 6 8 5 | 0 4. Answer each question.

1. How many students in total? _____
2. What is the lowest score? _____
3. What is the highest score? _____
4. What is the range? _____
5. How many scores are in the 40s? _____
6. What is the median? _____
7. How many scores are above 40? _____
8. What is the mode? _____
9. How many scores are below 35? _____
10. Write the scores in the 30s in order. _____

A dot plot shows: 0 (1 dot), 1 (2), 2 (2), 3 (3), 4 (2), 7 (1). Answer each question.

11. How many matches in total? _____
12. What is the mode? _____
13. What is the range? _____
14. How many matches had 2 goals? _____
15. How many matches had more than 3 goals? _____
16. What is the median? _____
17. What is the total number of goals? _____
18. What is the mean? _____
19. Which value is the outlier? _____
20. How many matches had fewer than 2 goals? _____

MEDIUM

12 QUESTIONS

A frequency table shows scores 0–6 with frequencies 3, 7, 6, 4, 2, 1, 1. Answer each question.

21. How many matches were played? _____
22. Find the range. _____
23. Find the mode. _____
24. Find the median. _____
25. Find the total number of goals (Σfx). _____
26. Find the mean (2 dp). _____
27. How many matches had 2 or fewer goals? _____
28. How many had at least 4 goals? _____
29. What percentage of matches had 1 goal? _____
30. What fraction had 0 goals? _____
31. How many more matches had 1 goal than 3 goals? _____
32. What is the highest score recorded? _____

Worksheet 4: Frequency tables and plots

Week 2 • Worksheet 4 of 6 • 40 questions • continued

NSW syllabus outcomes

MA4-DAT-C-01 (classifying and displaying data) • MA4-DAT-C-02
(measures of centre and range) • MAO-WM-01

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● CHALLENGING

8 QUESTIONS

Answer each question. Show your working.

1. 20 families own 3, 1, 2, 3, 4, 3, 2, 1, 4, 3, 3, 2, 4, 6, 3, 5, 4, 2, 3, 4 computers. Build a frequency table.

.....

2. For that data, what was the most frequent value?

.....

3. How many families owned 4 computers?

.....

4. What percentage of families owned one computer?

.....

5. For that data, find the mean number of computers (2 dp).

.....

6. Explain when a stem-and-leaf plot is more useful than a frequency table.

.....

7. A sector graph shows Maths 25%, English 20%, Science 30%, Art 15%, PE 10%. If 160 students were surveyed, how many chose English?

.....

8. For that sector graph, which subject is most popular and what percentage prefer Art?

.....

Worksheet 5: Reading and interpreting graphs

Week 3 • Worksheet 5 of 6 • 40 questions

NSW syllabus outcomes

MA4-DAT-C-01 (classifying and displaying data) • MA4-DAT-C-02
(measures of centre and range) • MAO-WM-01

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Always check the scale on the axes before reading values off a graph.

EASY

20 QUESTIONS

A bar graph shows pets: Dog 14, Cat 10, Fish 6, Bird 4, None 8. Answer each question.

1. How many students own a fish? _____
2. Which pet is the most common? _____
3. How many more own a dog than a bird? _____
4. How many own neither a dog nor a cat? _____
5. How many students were surveyed? _____
6. How many own a cat or a bird? _____
7. Which pet is the least common? _____
8. How many more own a cat than a fish? _____
9. What fraction own a dog? _____
10. What percentage own no pet? _____

A sector graph shows Maths 25%, English 20%, Science 30%, Art 15%, PE 10%. Answer each question.

11. Most popular subject? _____
12. Least popular subject? _____
13. Percentage who prefer Art? _____
14. If 160 were surveyed, how many chose English?

15. If 160 were surveyed, how many chose Science?

16. What percentage prefer Maths or English? _____
17. What fraction prefer PE? _____
18. How many degrees represent Science on the graph?

19. How many degrees represent PE? _____
20. If 80 were surveyed, how many chose Art? _____

MEDIUM

12 QUESTIONS

Answer each question about data displays.

21. Name the graph made of separated columns.

22. Name a graph that shows parts of a whole. _____
23. Which graph usually has time along the horizontal axis? _____
24. Which plot shows every individual value as a dot? _____
25. Which plot keeps the actual digits of the data?

26. Which graph is best for showing a trend over ten years? _____
27. Which display is best for finding the mode quickly?

28. Name a display suitable for categorical data. _____
29. Name a display suitable for numerical data. _____
30. What must every graph have besides the data?

31. Why should a bar graph have gaps between the bars?

32. What does the height of a bar represent? _____

Worksheet 5: Reading and interpreting graphs

Week 3 • Worksheet 5 of 6 • 40 questions • continued

NSW syllabus outcomes

MA4-DAT-C-01 (classifying and displaying data) • MA4-DAT-C-02
(measures of centre and range) • MAO-WM-01

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● CHALLENGING

8 QUESTIONS

Answer each question. Show your reasoning.

1. A line graph of yearly profit starts its vertical axis at 2 rather than 0. What impression does this create?

.....

2. Explain one other way a graph can be made misleading.

.....

3. Profits were 2.5, 2.25, 3.25, 3.5, 3.75 and 4 million over six years. Describe the trend.

.....

4. Why is a sector graph a poor choice for showing 20 categories?

.....

5. A survey of 500 people found 220 prefer coffee and 180 prefer tea. What percentage prefer neither?

.....

6. Draw a rough sketch showing what a fair version of the truncated profit graph would look like.

.....

7. A bar graph has bars of 14, 10, 6, 4 and 8. Explain how you would check the total is 42.

.....

8. Give one advantage and one disadvantage of a sector graph.

.....

Worksheet 6: Mixed revision

Week 3 • Worksheet 6 of 6 • 40 questions

NSW syllabus outcomes

MA4-DAT-C-01 (classifying and displaying data) • MA4-DAT-C-02
(measures of centre and range) • MAO-WM-01

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This sheet mixes every skill from the Analysing Data chapter. Work through it as though it were the topic test.

EASY

20 QUESTIONS

For the data set 5, 2, 3, 6, 4, 5, 8 find:

1. the mean _____
2. the median _____
3. the mode _____
4. the range _____

For the data set 6, 12, 11, 12, 10, 6, 6, 10, 6 find:

5. the mean _____
6. the median _____
7. the mode _____
8. the range _____

A bar graph shows pets: Dog 14, Cat 10, Fish 6, Bird 4, None 8.

9. How many own a fish? _____
10. Which pet is most common? _____
11. How many more own a dog than a bird? _____
12. How many own neither a dog nor a cat? _____

A sector graph shows Maths 25%, English 20%, Science 30%, Art 15%, PE 10%.

13. Most popular subject? _____
14. Percentage who prefer Art? _____
15. If 160 were surveyed, how many chose English?

16. Least popular subject? _____

A dot plot shows 0, 1, 1, 2, 2, 3, 3, 3, 4, 4, 7 goals.

17. What is the mode? _____
18. What is the outlier? _____
19. In how many matches were 2 goals scored? _____
20. What is the range? _____

MEDIUM

12 QUESTIONS

Answer each question.

21. A stem-and-leaf plot has 2 | 8, 3 | 2 5 9, 4 | 0 1 2 3 6
8, 5 | 0 4. How many students? _____
22. For that plot, what is the lowest score? _____
23. For that plot, what is the range? _____
24. For that plot, find the median. _____
25. The mean of 5 numbers is 12; four are 10, 11, 13, 15.
Find the fifth. _____
26. A data set has range 18 and smallest value 7. Find the
largest. _____
27. Find the mean of 20, 21, 23, 24, 25. _____
28. Find the median of 8, 4, 1, 1, 4, 1, 3, 6. _____
29. Find the mode of 12, 8, 12, 6, 12, 8. _____
30. Find the range of 45, 52, 48, 51. _____
31. Name the graph made of separated columns.

32. Name a graph that shows parts of a whole. _____



Worksheet 6: Mixed revision

Week 3 • Worksheet 6 of 6 • 40 questions • continued

NSW syllabus outcomes

MA4-DAT-C-01 (classifying and displaying data) • MA4-DAT-C-02
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● CHALLENGING

8 QUESTIONS

Answer each question. Show your working.

1. A small data set has a mean of 6. An extra value of 20 is added. Will the new mean be higher, lower or the same? Explain.
.....
2. A line graph of profit starts its vertical axis at 2 rather than 0. What is misleading, and what impression does it create?
.....
3. 20 families own 3, 1, 2, 3, 4, 3, 2, 1, 4, 3, 3, 2, 4, 6, 3, 5, 4, 2, 3, 4 computers. Build a frequency table.
.....
4. For that data, what percentage of families owned one computer?
.....
5. For that data, find the mean (2 dp).
.....
6. Explain why the median is not affected by an outlier but the mean is.
.....
7. Give a data set of five numbers with mean 10 and range 8.
.....
8. A survey of 500 found 220 prefer coffee, 180 prefer tea. Find the percentage who prefer neither.
.....